Homework 4: Asymptotis, Heap Implementation

1. $\log_2(n)$ is order n

Proof: For any positive integers c and n_0 there exists some integer $n \geq n_0$ such that $\log_2(n) < cn$. Let $n_0 = 1$ and $c = 1$

when $n = 1$ then $\log_2(1) \leq 1$

$= 0 \leq 1$ which is true

when $k \geq 1$ we can assume that $\log_2(k) \leq k$

we will show $\log_2(k+1) \leq k+1$

$k+1 \leq 2k$ for all $k \geq 1$

because $2k - (k+1) = k-1 \geq 0$

$\log_2(k+1) \leq \log_2(2k)$

$\log_2(2k) = \log_2(2) + \log_2(k) = 1 + \log_2(k)$

$1 + \log_2(k) \leq 1 + k$

$\log_2(k+1) \leq 1 + k = k+1$

So $\log_2(n) \leq n$ for all $n \geq 1$ thus $\log_2(n)$ is order n.

2. Prove 4^n is not order 2^n

Pf: There exists positive integers c and n_0 such that $n \geq n_0$ and $4^n > c2^n$

Choose $n = \max\{n_0, \log_2(c+1)\}$

$4^n = (2^2)^n = 2^{2n}$

$2^{2n} \leq c2^n$

$= 2^n \leq c$

$= 2^n \geq 2^{\log_2(c+1)} = c+1$

$= 2^n \geq c+1 > c$

$2^n \leq c$

therefore 4^n is not order 2^n

3. $n + n^2 + n^3$

$\log(n) + 2n$

$7n\log(n) + 6$

4^n

$10 \cdot 2^n + n^2$

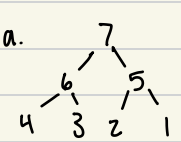
$n^3 + \log(n)$

$n\log(n) + \log(n)$

$2^n + n^{521}$

1. $\log(n) + 2n$
2. $7n\log(n) + 6$
3. $n\log(n) + \log(n)$
4. $n^3 + \log(n)$
5. $n + n^2 + n^3$
6. $10 \cdot 2^n + n^2$
7. $2^n + n^{521}$
8. 4^n

4a.



Index	Parent
1	0
2	0
3	1
4	1
5	2
6	2

4c.

Index	left child	right child
0	1	2
1	3	4
2	5	6

5. if each professor has 3 friends then to divide them into 2 rooms for each of the professors have them and 1 friend go to one room and the other 2 go to the other room. Then from there find out the other friend of the professors already assigned to the room and move the new professors so that only that professor and 1 friend is in the same room

professor	friends
~~1~~	~~2~~ ~~3~~ ~~4~~
~~2~~	~~1~~ 5 6
~~3~~	~~1~~ ~~4~~ 5
~~4~~	~~1~~ ~~3~~ ~~6~~
5	2 3 6
6	2 4 5

1 2 5	room1
3 4 6	room2